Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Symbiotic Probiotic Cosmeceutical for Acne Control and Skin Barrier Restoration			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio·Medical		Bio·Medical	
	Medium Category	Small Category	Medium Category	Small Category
	Industrial Biotechnology	Bio cosmetic / Materials	Industrial Biotechnology	Functional Biomaterials
Concept of Proposed Technology	The objective of this project is to develop a **next-generation probiotic cosmeceuticals** that simultaneously **controls acne, reduces inflammation, and restore skin barrier** using a **symbiotic microbiome-based approach** that integrates **probiotics and parabiotics.** Acne is the most prevalent skin disorder worldwide, yet conventional acne treatments frequently cause adverse effects such as skin dryness, microbiome imbalance, redness, delayed skin barrier recovery, antibiotic resistance, and increased risk of scarring. There remains a significant unmet need for acne solutions that are **highly effective, microbiome-respecting and gentle on the skin barrier**. The proposed technology addresses this gap by inhibiting *Cutibacterium acnes*, modulating inflammatory responses, and reinforcing skin barrier integrity in a coordinated manner, rather than focusing solely on antimicrobial effects. **Scientific Basis and Technical Focus** The project employs probiotic strains and parabiotics with previously verified for their ability to mitigate			

	acne and enhance skin barrier function. Building on existing clinical evidences, the study will be carried out to systematically verify—from a **skin microbiome perspective**—the mechanisms by which these probiotics contribute to inflammation reduction and barrier stabilization within the skin environment. To maximize functional efficacy, the project utilizes a **serum-infused sheet mask** as a representative product format. Due to its inherent characteristics, the sheet mask maintains close contact with the skin for a prolonged duration, facilitating direct and sustained interaction between functional ingredients and the skin surface microbiome. This makes it an ideal development platform for evaluating quantitative changes in inflammation and skin barrier function. The ultimate goal is to establish a **scientifically validated functional cosmetic technology** with highly targeted global commercialization potential.
Rationale for Joint R&D Project	Functional cosmetic technologies based on symbiotic microbiome represent a highly convergent field requiring integrated expertise in **microbe–skin interactions, mechanistic validation, and cosmetic formulation**. However, many probiotic cosmetic products currently on the market lack sufficient scientific validation of effectiveness, proven safety or mechanistic clarity, limiting their credibility. This project adopts a joint R&D framework in which: • The **Korean SME partner** is responsible for applying probiotics technologies to cosmetic formulations and developing product prototypes and to skin clinical trials; and • The **overseas research institute** conducts in-depth verification of anti-inflammation effects, barrier-related indicators, and microbiome-based mechanisms of action. Through this collaboration, the project ensures that research outcomes translate into practical, next-generation cosmeceutical products rather than remaining at a purely experimental level. By bridging microbiome-based biomaterials with functional cosmetic technologies, the project promotes convergence between the biotechnology and cosmetics industries and strengthens technological competitiveness for future global market expansion.
Key Words	1. Probiotics 2. Skin Microbiome, Symbiotic microbiome 3. Acne treatment & Skin Barrier 4. Functional Cosmetics, Cosmeceutical 5. Sheet mask cosmetic (Representative Application)

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (TRL 4)	**End of the Project**	Level (TRL 6)
R&D Objectives	The objective of this R&D project is to establish **probiotic-based skin microbiome technologies** as a scientifically validated approach for mitigating acne and restoration of skin barrier and to develop these technologies into high-performance cosmeceutical products. Specific objectives include:			

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	1. Analysis of changes in key skin inflammation, hydration and barrier-related indicators following application of probiotics and parapbiotics 2. Examination of correlations between skin microbiome modulation and improvements in acne control and skin barrier restoration 3. Design of functional cosmetic formulations that account for the biological and formulation characteristics of probiotics
R&D Target Specifications	• Confirmation of over 40% inflammation reduction indicator in comparing to the conventional acne topical treatments with 8-12% achieving "clear/almost clear" status in trials • Verification of changes in skin hydration associated with skin barrier stabilization by Tewameter (TEWL) method with 30-50% hydration level after 4-week application • Validation of inhibiting C.acnes and healthy microbiome balance by Laboratory analysis utilizing Culturing, Quantitative PCR (qPCR) and 16S rRNA Sequencing • Development of prototype-level acne management and skin barrier friendly functional cosmetic products
R&D Plan	**Phase 1: Review of Probiotic-Based Acne Management and Study Design** • Formulation of research hypotheses based on existing evidence related to acne controlling and barrier repair • Definition of key evaluation indicators for acne control and skin barrier function **Phase 2: Mechanistic Verification of Skin Microbiome and Acne Control** • Analysis of changes in acne-related and barrier-related indicators following probiotic application • Examination of correlations between microbiome composition changes and functional outcomes **Phase 3: Functional Cosmetic Formulation and Prototype Development** • Evaluation of formulation stability and sensory quality for probiotic application • Production of a serum-infused mask sheet prototype as a representative application case **Phase 4: Integration and Technology Maturation** • Consolidation of mechanistic and formulation results • Evaluation of potential expansion into additional cosmetic formulations

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Conduct in-depth experimental studies and data interpretation to provide scientific evidence of effectiveness and safety of probiotic acne controlling technology • Executing laboratory analysis to verify mechanisms of acne control and skin barrier improvement from a skin microbiome perspective • Provide scientific validation and advisory support based on research findings

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Expected R&R of Korean SME Partner	• Lead development of probiotic-based functional cosmetic formulations • Design and productize commercialization-ready cosmetic prototypes • Evaluate formulation performance including skin clinical trials of prototype and refine technology for commercialization • Define future product development and market expansion strategies

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Expected Outcomes** • Scientific validation of probiotic–based skin microbiome technologies • Development of prototype of acne-prone skin functional cosmetics using a serum-infused sheet mask • Validation of scientific efficacy and safety of acne control and skin barrier restoration of the probiotic acne controlling technologies **Industrial and Economic Impact** • Strengthening of competitiveness in microbiome-based functional cosmeceuticals • Creation of higher added value in acne care and functional cosmetic markets • Enhanced technological credibility to support global market expansion

6. References

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(3) Young-Sun Lee , Da-In Noh , Su-Jeong Lee , (2025), Anti-inflammatory and barrier-restoring effects of heat-killed Lactiplantibacillus plantarum postbiotics in an in vitro model of atopic dermatitis, Journal of Applied Microbiology.

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(5) Ho-Wei Lin , Ka-Wai Tam , Yu-Chen Huang, (2025) Efficacy of oral probiotics in patients with acne: a systematic review and meta-analysis of randomized trials Get access Arrow, Clinical and Experimental Dermatology,January 2026, Pages 68–77